Information for Candidates

Switch-Mode Power Supplies

Voltage ratio equations for buck SMPS $\frac{V_O}{V_I} = \delta$ or $\frac{V_O}{V_I} = \frac{1}{1+\frac{2fLI_O}{V_I\delta^2}}$

Voltage ratio equations for boost SMPS $\frac{V_O}{V_I} = \frac{1}{1-\delta}$ or $\frac{V_O}{V_I} = \frac{1}{1-\frac{V_I\delta^2}{2fLI_I}}$

Three-Phase Systems

Line Voltages and Current

Star $V_L = \sqrt{3}\,V_P;\ I_L = I_P$ Delta $V_L = V_P;\ I_L = \sqrt{3}\,I_P$

Power $P_{3\phi} = 3\,V_P I_P cos(\phi) = \sqrt{3}\ V_L I_L cos(\phi)$

Induction Machines

$\omega_S = {}^{\omega_E}/_{P}$; $s = \frac{\omega_S - \omega_R}{\omega_S}$; $T_{EM} = \frac{3\,I_R'^2\,R_R'}{\omega_R}\left(\frac{1-s}{s}\right)$

Photovoltaic Systems

$$I_{PV} = I_{Ph} - I_{AK} - I_{Sh} \qquad I_{AK} = I_0\left(e^{\frac{V_{PV}+I_{PV}R_S}{K_I v_T}} - 1\right)$$

Power Flow in Lines and Cables

Cable Parameters

$R'_{LF} = \frac{1}{\sigma_C \pi r_C^2} + \frac{1}{\sigma_C 2\pi r_O t_O}$ $L' = \frac{\mu}{2\pi} ln\left({}^{r_O}/_{r_C}\right)$ $C' = \frac{2\pi\varepsilon_0\varepsilon_{RI}}{ln\left({}^{r_O}/_{r_C}\right)}$ $G' = \frac{2\pi\sigma_I}{ln\left({}^{r_O}/_{r_C}\right)}$

OHL Parameters (approximate form)

$R'_{LF} = \frac{2}{\sigma_C \pi r_C^2}$ $L' = \frac{\mu}{\pi} ln\left({}^{d}/_{r_C}\right)$ $C' = \frac{\pi\varepsilon_0\varepsilon_{RI}}{ln\left({}^{d}/_{r_C}\right)}$ $G' = \frac{\pi\sigma_I}{ln\left({}^{d}/_{r_C}\right)}$

Power Flow (full form)

$$P_S = \frac{{V_S}^2}{Z_{SR}} cos(\theta) - \frac{V_R V_S}{Z_{SR}} cos(\theta+\delta) \qquad Q_S = \frac{{V_S}^2}{Z_{SR}} sin(\theta) - \frac{V_R V_S}{Z_{SR}} sin(\theta+\delta)$$

Voltage Drop (approximate form) $\Delta V = |V_S| - |V_R| \approx \frac{R\,P_S + X\,Q_S}{|V_S|}$

Answers

1. This question covers several topics and all parts should be attempted.

a) Consider a national-scale AC electricity system.

i) For an example country, the UK for instance, outline the changes in proportions of generation from various sources in the last decade and the reasons for the change. [4]

Discussion should note the rise of low-carbon / renewable sources, principally wind and photo-voltaic energy, in many countries of the world. [1 mark]

Rise of renewables is driven by subsidies in the form of feed-in tariffs, green certificates or contracts-for-difference, although subsidy-free invest is now viable in some cases. [1 mark]

Among the fossil fuels, gas has been displacing coal, partly driven by reducing gas prices (or expanding supply) and carbon taxes that increase the costs of coal more than gas. [1 mark]

Various other points can be made regarding position of nuclear, hydro or biomass. In the UK hydro is maintained and little prospect of expansion, biomass has been subsidised and is not substantial and new-build nuclear is subsidised to replace decommissioned plant. [1 mark]

ii) Explain why AC is used for almost all parts of an electricity system. [4]

Discussion should explain why a **variety** of voltages are needed such as EHV for bulk transmission and several lower voltages for local distribution [2 marks]. Then note that the transformer was is historically and still is a cost-effective (low capital cost and running cost) way of changing power at one voltage to another voltage but relies on a continuous rate of change of voltage, hence AC. [2 marks]

b) Consider a typical photovoltaic panel.

i) Explain the factors that mean that a photovoltaic panel does not produce its rated output continuously. [4]

[4 marks for full discussion of these factors]

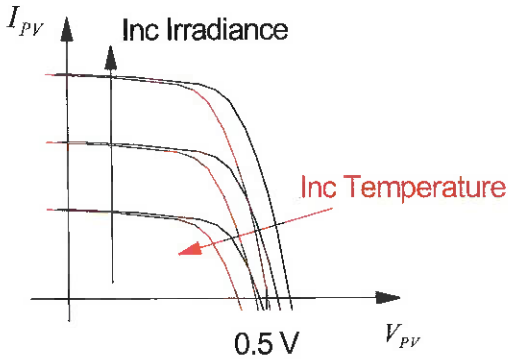
- Daily cycle of sun-rise and sun-set
- Season cycle of long and short days
- Angle of sun in the sky, which as latitude becomes lower in general and much lower in winter affecting both angle at which light impinges on panel (therefore effective use of its area) and absorption in the atmosphere from long atmospheric paths (higher air-mass)
- Cloud cover and shadowing from buildings and vegetation

ii) Sketch the shape of a current against voltage curve for a photovoltaic panel for two different temperatures and explain why the temperature dependency occurs. [4]

[2 marks diagram showing shapes like these]

[2 marks for explanation]

Answers



The current is close to constant at low voltages but decreases when the internal *pn* junction starts to forward bias and conduct. The junction current is set by

$$I_{AK} = I_0\left(e^{\frac{V_{PV}\,q}{K\,T}} - 1\right)$$

It is the reverse saturation current in this equation that increases strongly with temperature and thus the current available externally decreases with temperature.

c) Consider Buck and Boost SMPS.

i) State the assumptions that underlie the derivation of the expression for output voltage in continuous conduction mode, i.e., $\frac{V_O}{V_I} = \delta$. [4]

[4 marks for full discussion of these factors]

- The circuit is in periodic steady-state such that net change in inductor current over a cycle is zero.
- Mosfet and diode have negligible voltage drop while conducting.
- Inductor has no resistance
- Output capacitor is large valued and output voltage can be considered constant over a cycle.

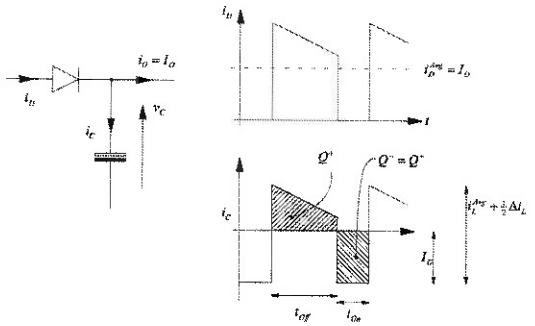
i) Draw the shape of the current that flows in the inductor, diode, and capacitor of the Boost SMPS for continuous operation. [4]

Inductor current comprises a triangular ripple around a dc term. [1 mark]

Diode current is a trapezoidal pulse train corresponding to the downward sloping portions of the inductor current [1 mark]

Capacitor current is the diode current off-set negatively by the (constant) output current. [2 marks]

Answers



iii) Explain why, for a similar selection of inductor, capacitor and switching frequency, a boost SMPS would be expected to have a higher output voltage ripple than a buck SMPS. You may consider the case of continuous conduction. [4]

The capacitor current shown in (ii) is a pulse train with a large peak to peak amplitude of $i_{c\,Boost}^{ptp} = I_L^{avg} + \frac{1}{2}\Delta i_L$. It is thus much larger in amplitude than the buck case which sees on the inductor ripple $i_{c\,Buck}^{ptp} = \Delta i_L$. The principal component of ripple are these terms multiplied by the ESR. So, for the same frequency and inductor, and similar voltages, the Δi_L terms are similar. For continuous conduction the I_L will dominate and boost will give the larger ripple voltage. [4 marks]

d) Consider overhead lines and cables

i) A single-phase overhead line has an impedance of 1 + j1.5 Ω and a sending-end voltage of 11 kV. It supplies a load of 18 + j1 Ω. Calculate the voltage at the load and the real and reactive power it consumes. [4]

[1 mark for potential divider rule correctly applied (or alternative calculation via finding current]

$$V_L = V_S \frac{Z_L}{Z_L + Z_{Line}} = 11 \times 10^3 \frac{18 + j1}{18 + j1 + 1 + j1.5} = 11 \times 10^3 \frac{18.03\angle 3.18°}{19.16\angle 7.50°}$$

$$= 10.35\ kV \angle -4.32°$$

[3 marks accurate answers for P and Q and for units.]

$$I_L = \frac{V_S}{Z_L + Z_{Line}} = \frac{11 \times 10^3}{18 + j1 + 1 + j1.5} = \frac{11 \times 10^3}{19.16\angle 7.50°} = 574\ A \angle -7.50°$$

$$S = V_L I_L^* = 10.35 \times 10^3 \angle -4.32° \times 574\ A \angle +7.50°$$

$$P = Re\{V_L I_L^*\} = 5.93\ MW$$
$$Q = Re\{V_L I_L^*\} = 0.33\ MVAr$$

Answers

ii) The same line and sending-end voltage are used to supply a constant power load of 8 MW and 1 MVAr. Estimate the difference in voltage magnitude between the sending- and receiving-end voltages. [4]

[1 mark for recognising approximate equation need]

[1 mark for further approximation of sending-end power by receiving-end power.]

[2 marks for accurate calculation]

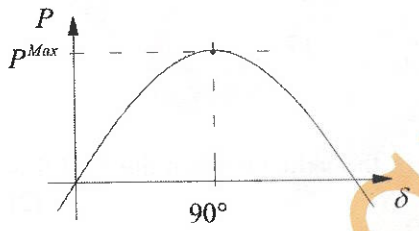
$$\Delta V = |V_S| - |V_R| \approx \frac{R\,P_S + Z\,Q_S}{|V_S|}$$

$$P_S \approx P_R \quad Q_S \approx Q_R$$

$$\Delta V \approx \frac{R\,P_R + Z\,Q_R}{|V_S|} = \frac{1.0 \times 8 \times 10^6 + 1.5 \times 1 \times 10^6}{11 \times 10^3} = 863\,V$$

iii) Sketch the shape of the graph of power against load angle (angle difference between sending- and receiving-end voltages) for an overhead line that is inductance dominated and use this to explain the angle limit on power that a line can transmit. [4]

[2 marks for graph; 2 marks for discussing the need to keep angle below 90°]



Graph is based on approximation of $P_S = \frac{V_S^2}{Z_{SR}}cos(\theta) - \frac{V_R V_S}{Z_{SR}}cos(\theta + \delta)$ when impedance dominated which is $P = \frac{V_R V_S}{X_{SR}}sin(\delta)$. Maximum power occurs at an angle of 90°. In practice an angle somewhat less is set as the maximum to allow for transient conditions without a risk of power decreasing, not increasing with angle, leading to instability.

Answers

2.

A **boost** switch-mode power supply (SMPS) is to be used to convert 25 V to 100 V. The SMPS is intended to operate at a maximum output current of 4 A.

The MOSFET to be used has an on-state resistance of $R_{DS(on)} = 50$ mΩ and turn-on and turn-off energy losses of $E_{Sw\text{-}on} = 55$ μJ and $E_{Sw\text{-}off} = 35$ μJ.

The diode to be used has an on-state voltage of $V_{AK(on)} = 0.95$ V.

The MOSFET is attached to a heatsink with a thermal resistance (sink to air) of 2.5 K/W, the MOSFET itself has a thermal resistance (junction to case) of 0.4 K/W and the mounting washer has a thermal resistance (case to sink) of 0.1 K/W. An ambient temperature of 40°C will be considered and the junction temperature should not exceed 100°C

You may assume that the SMPS operates in continuous conduction mode for the questions that follow.

a)

i) Calculate the duty-cycle with which the SMPS should operate. [2]

Apply standard formula [2 marks for accurate answer].

$$\delta = \frac{V_O - V_I}{V_O} = \frac{100 - 25}{100} = 0.75$$

iii) Calculate the average value of the current in the inductor when the SMPS is operating at its maximum output current. [2]

Apply standard formula [2 marks for accurate answer].

$$I_L = I_I = I_O \frac{V_O}{V_I} = 4\frac{100}{25} = 16\,A$$

b)

i) Calculate the maximum switching frequency with which the MOSFET can be operated without exceeding its maximum permissible junction temperature. [8]

First find maximum allowed power loss from thermal conditions [2 marks for equation; 2 for accurate answer]

$$P_{Loss}^{Max} = \frac{\theta_J - \theta_A}{R_{Th-JC} + R_{Th-CS} + R_{Th-SA}} = \frac{100 - 40}{0.4 + 0.1 + 2.5} = 20\,W$$

Then find maximum frequency from power loss equation [2 marks for equation; 2 for accurate answer]

$$f = \frac{P_{Loss}^{Max} - \delta I_{DS}^2 R_{DS}}{R_{On} + R_{Off}} = \frac{20 - 0.75 \times 16^2 \times 0.05}{55 \times 10^{-6} + 35 \times 10^{-6}} = 115.5\,kHz$$

ii) Calculate a value for inductor such that the ripple current in the inductor will be 0.5 A when operated at the frequency found in (i). [4]

Answers

Re-arrange equation for current ripple to find inductance. [2 marks for equation; 2 for correct answer]

$$L = \frac{V_I}{\Delta i_L} \times \frac{\delta}{f} = \frac{25 \times 0.75}{0.5 \times 115.5 \times 10^3} = 325\ \mu H$$

iii) Assume that the values found so far are used and choose a capacitor such that the voltage ripple components across its capacitance and effective series resistance are 20 mV and 100 mV respectively when operating at full output current. [4]

[2 marks for equations; 2 for accurate answers]

$$C = \frac{I_O(1-\delta)}{f\,\Delta v_C} = \frac{4 \times (1-0.75)}{115.5 \times 10^3 \times 20 \times 10^{-3}} = 430\ \mu F$$

$$R_{ESR} = \frac{\Delta v_{ESR}}{I_I + \tfrac{1}{2}\Delta i_L} = \frac{0.1}{16 + \tfrac{1}{2} \times 0.5} = 6.1\ m\Omega$$

c)

i) The control circuit of the SMPS is known to consume 1.25 W. Calculate the power losses in the MOSFET and diode (you may ignore losses in the resistance of the inductor and capacitor) for operation at maximum output power and hence calculate the efficiency of the SMPS at this point. [4]

[2 marks for equation (some re-use from (b); 2 for accurate answers]

$$P_{Mos\ cond} = \delta\, I_I^2\, R_{DS} = 0.75 \times 16^2 \times 0.050 = 9.60\ W$$

$$P_{Mos\ Sw} = f \times (E_{on} + E_{off}) = 115.5 \times 10^3 \times 90 \times 10^{-6} = 10.40\ W$$

$$P_{D\ cond} = (1-\delta)\, I_I\ \ V_{AK} = 0.25 \times 16 \times 0.95 = 3.80\ W$$

$$\eta = \frac{V_O I_O}{V_O I_O + P_{Control} + P_{Mos\ cond} + P_{Mos\ Sw} + P_{D\ cond}} = \frac{4 \times 100}{4 \times 100 + 1.5 + 9.60 + 10.40 + 3.80}$$

$$\eta = 94.1\ \%$$

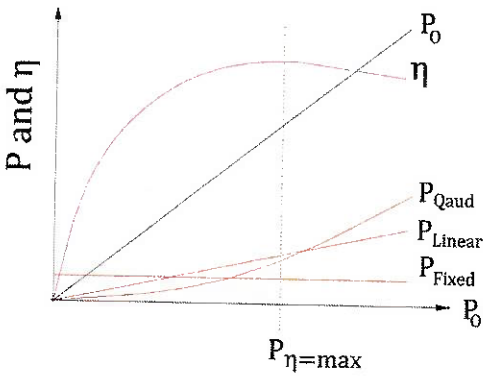
ii) Sketch a graph of how efficiency is expected to change with output power and include how each of the power loss terms scale with output power (you should assume that the switching energy losses, E_{on}, E_{off} , scale linearly with current). [3]

One term is fixed: control circuit loss of 1.5W

Two terms are linear: diode conduction loss: Mosfet switching loss.

One term is quadratic: Mosfet conduction loss

Answers



(iii) Treat the power loss terms as a quadratic with coefficients a, b and c, and find the output power at which efficiency is maximum in terms of these coefficients and hence calculate this power. [3]

[1 mark for using quotient rule to identify maxima, 1 mark for correctly identifying coefficients and 1 mark for calculating final answer]

$$\eta = \frac{P_O}{P_O + a + bP_O + cP_O^2}$$

$$\frac{d\eta}{dP_O} = \frac{(a + (1+b)P_O + cP_O^2) - P_O\left((1+b) + 2cP_O\right)}{(a + (1+b)P_O + cP_O^2)^2}$$

Set $\frac{d\eta}{dP_O} = 0$ to find extrema. Only numerator can make this zero.

$$(a + (1+b)P_O + cP_O^2) - \left((1+b)P_O + 2cP_O^2\right) = 0$$

$$P_O^2 = \frac{a}{c}$$

Identify the coefficients by comparison to loss equations (b is shown here for completeness but is not required).

$$a = P_{Control}$$

$$P_{Linear} = f \times \left(E_{on} + E_{off}\right)\frac{I_l}{I_l^{Max}} + (1-\delta)\, I_l\ \ V_{AK} \therefore b = \left\{f \times \left(E_{on} + E_{off}\right) + (1-\delta)\ V_{AK}\right\}\frac{1}{P_O^{Max}}$$

$$P_{Mos\ cond} = \delta\, I_l^2\, R_{DS} = \frac{\delta\, P_O^2\, R_{DS}}{V_l^2} \therefore c = \frac{\delta\, R_{DS}}{V_l^2}$$

$$P_O^2 = \frac{a}{c} = \frac{P_{Control}}{\delta\, R_{DS}/V_l^2} = \frac{1.5 \times 25^2}{0.75 \times 0.05} = 25{,}000$$

$$P_O = \sqrt{25{,}000} = 158\ W$$

Answers

© Imperial College London

3. This question addresses several aspects of induction machines.

a)

i) The rotor of an induction machine has no electrical connection to an external supply and no permanent magnets. Explain why this is a desirable feature and also explain how it is that torque is exerted on the rotor. [6]

[1 mark for each point]

- Electrical connection to a rotating part via brushes gives rise of a need for frequency maintenance.
- Rubbing contacts can be a source of electromagnetic emissions and an ignition source because of the breaking of a path by the mechanical contracts.
- Permanent magnets do not scale up to high powers cost-effectively, are difficult to machine and shape and have a relatively low temperature limit.
- An induction machine has voltages induced on in the rotor conductors because there is a rate of change of flux linkage with them.
- It is the relative velocity between the rotation of the stator field and the bars themselves that determines the amplitude of induced voltage and hence current flow.
- Currents flowing as a result of the induced voltage interact with the stator field to create torque (such as to oppose the relative motion)

ii) What are the causes of inefficiency in an induction machine and how are they represented in the equivalent circuit. [4]

[1 mark for each point]

There are four major power loss mechanisms in an induction machine.

- Joule (conductive heating) loss in the resistance of the stator winding represented as a resistance, R_S, in series in the stator branch.
- Magnetic core losses, both eddy current heating loss and hysteresis loss, represented approximately as a resistance, R_M in parallel in the magnetising branch.
- Joule (conductive heating) loss in the resistance of the rotor winding represented as a referred resistance, R'_R, in series in the rotor branch.
- Mechanical losses caused by bearing friction and air-drag on the rotor (and wafter/fan) known as windage and friction losses. These are not represented in the equivalent circuit and must be accounted for after calculation of mechanical power or torque.

b) A three-phase induction machine with 3 pole-pairs has the following equivalent circuit parameters:

Stator resistance, $R_S = 8.0\ \Omega$.

Stator leakage reactance, $X_S = 20\ \Omega$;

Magnetising resistance, $R_M = 5000\ \Omega$;

Magnetising reactance, $X_M = 500\ \Omega$;

Referred rotor leakage reactance, $X_R = 20\ \Omega$;

Referred rotor resistance, $R_R = 6.4\ \Omega$;

ANSWERS

The machine is supplied at a phase voltage of 3,300 V and a frequency of 50 Hz. When used to drive a particular mechanical load, the machine runs at 960 rpm.

i) Calculate the slip. [2]

Slip is calculated from: $s = \frac{\omega_S - \omega_R}{\omega_S}$

$$\omega_S = {}^{\omega_E}/_{P} = 50 \times 2\pi/3 \frac{\text{rad}}{\text{s}} or\ 1{,}000\ rpm$$

$$s = \frac{\omega_S - \omega_R}{\omega_S} = \frac{1{,}000 - 960}{1{,}000} = 0.04$$

ii) Give an expression for the total input impedance of the machine and calculate its value in complex form. [6]

[3 marks for correct formulation of impedance, including correct parallel terms and inclusion of s]

Total impedance:

$$Z_T = R_S + jX_S + \left(R_M // jX_M // \left(\frac{R'_R}{s} + jX'_R\right)\right)$$

[3 marks for accurate computation]

$$Z_T = 140.3 + j77.6 = 160.3\angle 28.97°\ \Omega$$

iii) Calculate the electro-magnetic torque. [8]

First, find stator current from the phase voltage $I_S = \frac{V_S}{Z_T} = \frac{3300}{160.3\angle 28.97°} = 20.58\angle -28.97°\ \Omega$

[2 marks for accurate answer]

Then rotor current is found from the current divider rule (or otherwise)

$I_R = I_S \times \left(\frac{R_M // jX_M}{R_M // jX_M + \left(\frac{R'_R}{s} + jX'_R\right)}\right)$

[1 mark for correct formulation]

$I_R = 18.42\ \angle$-12.54° [1 mark for accurate answer]

Power from standard formula $P_{EM} = 3\, I'^2_R\, R'_R \left(\frac{1-s}{s}\right) = 3 \times 18.41^2 \times 6.4 \times \left(\frac{1-0.04}{0.04}\right) = 156.3$ kW

[1 mark for correct formula; 1 mark for accurate answer]

Torque found by dividing by rotor speed

$$T = \frac{P_{EM}}{\omega_R} = \frac{156{,}300}{(1 - 0.04) \times \frac{100\pi}{3}} = 1.55\ \text{kNm}$$

[1 mark for correct formula; 1 mark for accurate answer]

Answers

iv) Calculate the efficiency. [4]

[2 marks for method; 2 marks for accurate answer]

Take mechanical power from (iii) (or calculate there is torque was found directly in (iii)).

Calculate input power

$$P_{In} = 3V_S \times I_S \times cos(\angle Z) = 3 \times 3{,}300 \times 20.58 \times cos(28.97°) = 178.3\ kW$$

$$\eta = \frac{P_{Net}}{P_{In}} \times 100\% = \frac{156.3k}{178.3k} \times 100\% = 87.68\%$$

Answers